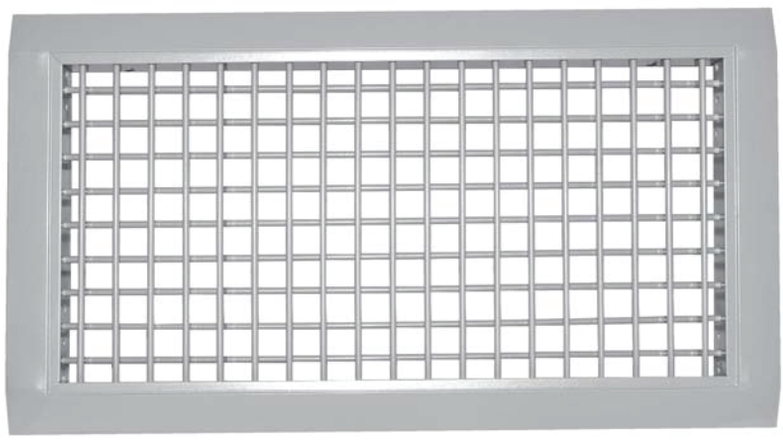
Ventilation Grille for Circular Air Ducts RDD/RDD1



RDD

Description
Rectangular aluminum grille for mounting on circular air ducts.

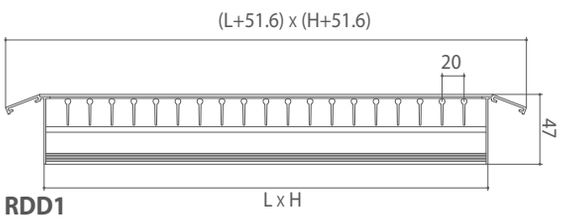
Application
Supply/ exhaust of the air flow.

Characteristics
The grille is equipped with individually adjustable blades and countersinking holes for mounting.
Movable blades allow airflow direction in desired areas.
RDD has two rows of adjustable blades and **RDD1** has a single row of blades.
Grids over 500 mm in length are provided with intermediate reinforcement rods.
The air flow can be adjusted by mounting a air control damper **OBD** on the grille's inlet.

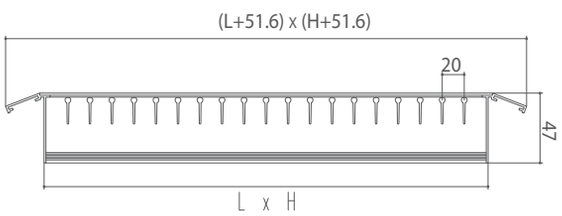
Accessories
Air control damper with opposable blades **OBD.**

Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016
Other RAL colors available on request

Technical design
RDD



RDD1



Grill Dimensions	Ø tube	
	min.	max.
425 x 75	160	400
525 x 75	160	400
625 x 75	160	400
425 x 125	315	900
525 x 125	315	900
625 x 125	315	900
425 x 225	630	1400
525 x 225	630	1400
625 x 225	630	1400
825 x 225	630	1400
1025 x 225	630	1400

Assembly Method
Standard mounting, with screws

Selection table RDD/RDD1

Air flow m³/h	LxH	425 x 75	525 x 75	625 x 75	425 x 125	525 x 125	625 x 125	425 x 225	525 x 225	625 x 225	825 x 225	1025 x 225
	Ak[m²]	0.013	0.016	0.019	0.025	0.031	0.037	0.049	0.061	0.073	0.097	0.122
200	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	5.1 24.0 4.3 12.1	4.6 20.0 3.5 8.0									
250	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	6.4 29.0 5.3 18.8	5.7 25.0 4.3 12.4	5.3 22.0 3.7 8.8								
300	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	7.6 33.0 6.4 27.1	6.9 29.0 5.2 17.9	6.3 26.0 4.4 12.7	5.5 21.0 3.3 7.3							
350	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	8.9 37.0 7.5 36.9	8.0 33.0 6.1 24.4	7.4 30.0 5.1 17.3	6.4 24.0 3.9 10.0	5.8 20.0 3.1 6.5						
400	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	10.2 40.0 8.5 48.2	9.2 36.0 6.9 31.8	8.4 33.0 5.8 22.6	7.3 28.0 4.4 13.0	6.6 23.0 3.6 8.5	6.0 20.0 3.0 6.0					
450	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	11.5 43.0 9.6 61.0	10.3 39.0 7.8 40.3	9.5 36.0 6.6 18.6	8.3 31.0 5.0 16.5	7.4 26.0 4.0 10.7	6.8 23.0 3.4 7.5					
500	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]		11.5 42.0 8.7 49.7	10.5 38.0 7.3 35.3	9.2 33.0 5.6 20.4	8.2 29.0 4.5 13.2	7.6 25.0 3.8 9.3	6.6 20.0 2.8 5.3				
600	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]			12.6 43.0 8.8 50.8	11.0 38.0 6.7 29.3	9.9 33.0 5.4 19.1	9.1 30.0 4.5 13.4	7.9 24.0 3.4 7.6	7.1 20.0 2.7 4.9			
700	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]				12.9 41.0 7.8 39.9	11.5 37.0 6.3 26.0	10.6 34.0 5.3 18.2	9.2 28.0 4.0 0.4	8.2 24.0 3.2 6.7	7.5 20.0 2.7 4.7		
800	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]				14.7 45.0 8.9 52.1	13.2 40.0 7.2 33.9	12.1 37.0 6.0 23.8	10.5 31.0 4.5 3.6	9.4 27.0 3.6 8.8	8.6 24.0 3.0 6.1		
900	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]					14.8 43.0 8.1 42.9	13.6 40.0 6.8 30.1	11.8 34.0 5.1 17.2	10.6 30.0 4.1 11.1	9.7 26.0 3.4 7.7	8.4 21.0 2.6 4.4	
1000	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]						15.1 42.0 7.5 37.2	13.1 37.0 5.7 21.2	11.8 33.0 4.6 13.7	10.8 29.0 3.8 9.6	9.3 23.0 2.9 5.4	
1200	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]							15.7 41.0 6.8 30.5	14.1 37.0 5.5 19.7	12.9 33.0 4.6 13.8	11.2 28.0 3.4 7.8	10.0 23.0 2.7 4.9
1400	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]							18.4 45.0 7.9 41.6	16.5 41.0 6.4 26.8	15.1 37.0 5.3 18.7	13.1 32.0 4.0 10.6	11.6 27.0 3.2 6.7
1600	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]								18.8 44.0 7.3 35.0	17.2 40.0 6.1 24.5	14.9 35.0 4.6 13.9	13.3 30.0 3.6 8.8
1800	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]									19.4 43.0 6.8 31.0	16.8 38.0 5.2 17.5	15.0 33.0 4.1 11.1
2000	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]										18.7 40.0 5.7 21.6	16.6 36.0 4.6 13.7
2500	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]											20.8 41.0 5.7 21.4

Vk = 0.2 (m/s)

NR < 25 | 25≤NR≤35 | NR>35

Annotation

* Ak [m²] - Free surface
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* Vk [m/s] - Velocity at which the air jet length was calculated "x" [m]

Order codes

Type + **OBD** / **Dimensions** / **L x H** / **0**

RDD / RDD1

Air control damper

L x H

425 x 75
525 x 75
625 x 75

425 x 125
525 x 125
625 x 125

425 x 225
525 x 225
625 x 225
825 x 225
1025 x 225

Other dimensions on request

0 Painted white RAL 9016
R Other RAL paints